
◆ 1. EC2 (Elastic Compute Cloud)

Summary:

EC2 provides **resizable compute capacity** — i.e., virtual machines in the cloud. You can launch, configure, and manage Linux/Windows instances.

Interview Q&A:

Q1: What is EC2 used for? A1: To run servers, host applications, deploy containers, run background jobs, etc.

Q2: What's the difference between EC2 and Lambda? A2:

- EC2 = long-running servers you manage
- Lambda = event-driven, no server management

Q3: What are EC2 instance types? A3:

- General purpose (t3, t4g)
- Compute optimized (c5, c6g)
- Memory optimized (r5, r6g)
- Storage optimized (i3)
- GPU (p3, g4)

Q4: How do you secure EC2 instances? A4:

- Use Security Groups, Key Pairs, IAM roles, and VPCs
-

◆ 2. S3 (Simple Storage Service)

Summary:

S3 is a **highly durable, scalable object storage** system for files, backups, logs, images, etc. It provides storage via buckets and supports versioning, encryption, lifecycle rules.

Interview Q&A:

Q1: What is S3 used for? A1: To store and retrieve data like media files, logs, backups, static websites.

Q2: What is the durability of S3? A2: 99.999999999% (11 9s) durability — highly reliable.

Q3: Can you host a static website on S3? A3:  Yes. Enable static website hosting on the bucket and set index/error files.

Q4: How do you control access to an S3 bucket? A4:

- Bucket policies
 - IAM policies
 - ACLs
 - Object-level encryption
-

3. Lambda

Summary:

Lambda lets you **run code without provisioning servers**. It's **event-driven**, scalable, and cost-efficient for micro-tasks, APIs, or backend jobs.

Interview Q&A:

Q1: What is AWS Lambda? A1: A serverless compute service where you upload code, define a trigger (e.g., API Gateway, S3, CloudWatch), and AWS handles the execution.

Q2: Use case example? A2:

- Resize images on S3 upload
- REST APIs (with API Gateway)
- Scheduled cron jobs

Q3: How is Lambda billed? A3: Based on:

- Number of requests
- Duration (in ms)
- Memory allocated

Q4: What is the max timeout and memory for a Lambda function? A4:

- Timeout: 15 minutes
 - Memory: 10 GB
-

◆ 4. IAM (Identity & Access Management)

Summary:

IAM controls **who can access what** in AWS. It lets you manage **users, roles, groups, policies**, and define **permissions at a granular level**.

Interview Q&A:

Q1: What is IAM? A1: IAM is the security layer of AWS used to manage **users, roles, and permissions** for accessing AWS resources.

Q2: What's the difference between user and role? A2:

- **User:** assigned to a real person
- **Role:** assumed temporarily by services or users (e.g., EC2 assuming an S3 role)

Q3: What is an IAM policy? A3: A JSON document that defines allowed/denied actions on AWS resources.

Q4: How to follow the principle of least privilege? A4:

- Grant only the permissions required
 - Use managed or customer policies
 - Monitor via Access Analyzer
-
-

◆ 5. RDS (Relational Database Service)

Summary:

RDS is a managed database service for running relational DBs like **MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server**.

It automates:

- Backups
 - Patching
 - Monitoring
 - High availability (via Multi-AZ)
-

Interview Q&A:

Q1: What is RDS used for? A1: Hosting managed relational databases without worrying about infrastructure management.

Q2: What are RDS deployment options? A2:

- **Single AZ:** Simple, lower cost
- **Multi-AZ:** High availability and failover support

Q3: What is the difference between RDS and Aurora? A3:

- **RDS:** Traditional DB engines
- **Aurora:** AWS-native, MySQL/PostgreSQL compatible, better performance (up to 5x)

Q4: Can RDS auto-scale? A4: Storage can auto-scale. Compute scaling must be manual or via Aurora Serverless (v2).

6. CloudWatch

Summary:

CloudWatch is AWS's observability service for **logs, metrics, dashboards, and alarms**.

Used to monitor:

- EC2, Lambda, RDS, S3, and custom metrics
 - Log storage and querying via CloudWatch Logs
 - Alerting via Alarms + SNS
-

Interview Q&A:

Q1: What can you monitor using CloudWatch? A1:

- CPU, disk, memory
- Lambda invocation metrics
- Custom app metrics
- Log errors & patterns

Q2: How do CloudWatch Alarms work? A2: You define thresholds on metrics, and trigger actions (like SNS, Lambda, EC2 scale-out) when crossed.

Q3: Difference between CloudWatch Logs and Metrics? A3:

- **Metrics:** Numeric data over time
- **Logs:** Raw text logs (like stdout, error messages)

Q4: Can you create custom metrics? A4:  Yes, using `PutMetricData` via SDK/CLI.

7. CloudFormation

Summary:

CloudFormation is an **Infrastructure as Code (IaC)** tool to define AWS resources in **YAML or JSON templates**.

Used to:

- Create/tear down full stacks
 - Automate repeatable infrastructure deployments
-

Interview Q&A:

Q1: What is AWS CloudFormation? A1: It's a service for defining and provisioning AWS infrastructure using code.

Q2: What is a CloudFormation Stack? A2: A deployed unit of your CloudFormation template — a collection of AWS resources managed as a single unit.

Q3: Example resource snippet (YAML):

Resources:

MyBucket:

Type: AWS::S3::Bucket

Q4: Benefits of CloudFormation? A4:

- Version control for infra
- Repeatable deployments
- Automated rollback
- No manual config drift

◆ 8. CloudTrail

Summary:

CloudTrail records **API activity across AWS accounts**. It logs who did what, from where, and when.

Used for:

- Auditing
- Security forensics
- Compliance

Interview Q&A:

Q1: What does CloudTrail do? A1: It captures every API call made in your AWS account — by CLI, SDK, or Console.

Q2: What's the difference between CloudTrail and CloudWatch? A2:

- **CloudTrail:** Logs **actions** (e.g., who deleted an S3 bucket)
- **CloudWatch:** Logs **system activity and performance** (CPU, logs)

Q3: Where are CloudTrail logs stored? A3: In an S3 bucket you configure. You can also forward to CloudWatch Logs.

Q4: Can CloudTrail be used across accounts? A4:  Yes, via **organization trails** in AWS Organizations.

◆ 9. SNS / SQS

Summary:

- SNS (Simple Notification Service) is a **pub/sub** messaging service to push messages to multiple subscribers (email, SMS, Lambda, SQS).
 - SQS (Simple Queue Service) is a **message queue** service for **decoupling microservices**.
-

Interview Q&A:

Q1: What is the difference between SNS and SQS? A1:

- SNS → Push-based, broadcast to many
- SQS → Pull-based, one message consumed by one service

Q2: Use cases for SNS? A2:

- Sending alerts
- Notifying multiple systems (email + Lambda + SQS)

Q3: Use cases for SQS? A3:

- Decoupling producers and consumers
- Handling async job queues

Q4: Types of SQS queues? A4:

- **Standard**: high throughput, at-least-once delivery
- **FIFO**: strict ordering and exactly-once delivery

Q5: Can SNS send to SQS? A5:  Yes — you can create fan-out architecture by publishing SNS messages that trigger SQS queues.

◆ 10. Cognito

Summary:

Cognito provides user authentication, authorization, and user management for web and mobile apps.

Supports:

- User sign-up/sign-in (User Pools)
 - Federated identity (Google, Facebook, SAML)
 - Temporary AWS credentials (Identity Pools)
-

Interview Q&A:

Q1: What is Amazon Cognito used for? A1: To manage user authentication, tokens, social logins, and access control in AWS-integrated apps.

Q2: What are User Pools vs Identity Pools? A2:

- **User Pool:** User directory, handles sign-in/sign-up
- **Identity Pool:** Grants temporary AWS credentials to users

Q3: Can Cognito integrate with third-party providers? A3:  Yes — supports Google, Facebook, SAML, OpenID Connect.

Q4: How does Cognito issue tokens? A4: Via **JWT** tokens — Access, ID, and Refresh tokens.

11. Route 53

Summary:

Route 53 is AWS's **DNS service** for registering domains, routing traffic globally, and performing health checks.

Supports:

- A, CNAME, Alias records
 - Latency-based routing
 - Failover
 - Weighted and geolocation routing
-

Interview Q&A:

Q1: What does Route 53 do? A1: It maps domain names to AWS resources (EC2, S3, CloudFront) or external IPs.

Q2: What is an Alias record? A2: AWS-specific DNS record that allows mapping to AWS resources (e.g., S3, CloudFront) without needing IP addresses.

Q3: Can Route 53 do health checks? A3:  Yes. You can configure health checks to reroute traffic in case of failures.

Q4: What is latency-based routing? A4: Directs users to the region that gives the **lowest latency**, improving performance.

◆ 12. Cost Explorer

Summary:

Cost Explorer lets you analyze, visualize, and forecast your **AWS usage and spending**. Useful for:

- Budgeting
 - Identifying cost spikes
 - Optimizing resources
-

Interview Q&A:

Q1: What is AWS Cost Explorer used for? A1: To view and analyze your spending by service, account, tags, and time range.

Q2: Can it help forecast future costs? A2:  Yes, it provides predictions based on past usage trends.

Q3: How do tags help in Cost Explorer? A3: Tagging resources allows you to group, filter, and analyze costs per team, project, or environment.

Q4: What's the difference between Cost Explorer and Budgets? A4:

- **Cost Explorer:** Historical and forecast reports
 - **Budgets:** Alerts when thresholds are breached
-
-

◆ 13. CloudFront

Summary:

CloudFront is AWS's **Content Delivery Network (CDN)** — it caches and delivers content (HTML, JS, images, video) from **edge locations** close to users, reducing latency.

Supports:

- Static and dynamic content
 - HTTPS
 - Geo-restriction
 - Signed URLs/tokens for private content
-

Interview Q&A:

Q1: What is CloudFront used for? A1: To distribute content globally with low latency and high transfer speed using edge caching.

Q2: What are edge locations in CloudFront? A2: Globally distributed data centers where content is cached and served to nearby users.

Q3: What's the difference between CloudFront and S3? A3:

- S3 stores the data
- CloudFront distributes it efficiently worldwide

Q4: How do you secure private content? A4: Using **signed URLs** or **signed cookies** with origin access control.

◆ 14. ELB (Elastic Load Balancer)

Summary:

ELB automatically distributes incoming traffic across multiple targets (EC2, containers, Lambda). Types:

- ALB (Application Load Balancer)
 - NLB (Network Load Balancer)
 - CLB (Classic — legacy)
-

Interview Q&A:

Q1: What are the types of ELBs and when to use them? A1:

- **ALB:** HTTP/HTTPS traffic, URL-based routing
- **NLB:** TCP/UDP, low latency
- **CLB:** Deprecated, basic Layer 4/7 load balancer

Q2: Can ELB work with auto scaling? A2:  Yes. It automatically distributes traffic among healthy instances in the Auto Scaling Group.

Q3: Does ELB support SSL termination? A3:  Yes — especially with ALB, SSL certs can be managed via ACM.

Q4: What is target group in ALB? A4: A logical group of resources (EC2, Lambda, IPs) to route traffic to, based on path/rules.

◆ 15. Budgets

Summary:

AWS Budgets lets you set custom cost and usage budgets and get alerts when thresholds are crossed.

You can track:

- Monthly cost
 - Specific service usage
 - Reserved Instance coverage
 - Savings Plan utilization
-

Interview Q&A:

Q1: How are Budgets different from Cost Explorer? A1:

- **Budgets:** Proactive alerting when thresholds exceed
- **Cost Explorer:** Historical and forecast data

Q2: Can budgets trigger actions? A2:  Yes — via SNS notifications or automation through Lambda.

Q3: Can budgets be tied to specific tags? A3:  Yes, for tracking spend by team, project, etc.

Q4: Budget example? A4: "Alert me if S3 costs exceed \$100 this month."

◆ 16. S3 Lifecycle

Summary:

S3 Lifecycle Policies help **automate data transition and expiration**. You can move objects to cheaper storage classes (e.g., Glacier) or delete them after a period.

Interview Q&A:

Q1: What is an S3 Lifecycle rule? A1: A policy that automatically transitions objects between storage classes or deletes them after a set period.

Q2: Common storage class transitions? A2:

- Standard → Infrequent Access → Glacier → Glacier Deep Archive

Q3: Can you apply lifecycle policies to specific prefixes/tags? A3:  Yes, rules can be scoped to object prefixes or tags.

Q4: Use case for lifecycle policy? A4: Archive logs to Glacier after 30 days; delete them after 365 days.

◆ 17. Auto Scaling

Summary:

Auto Scaling automatically adjusts the number of EC2 instances (or ECS tasks, DynamoDB throughput, etc.) based on demand.

You define:

- **Scaling policies** (target tracking, step scaling, etc.)
 - **Minimum/maximum instance limits**
 - **Health checks and replacement logic**
-

Interview Q&A:

Q1: What is Auto Scaling in AWS? A1: It automatically increases or decreases the number of resources (e.g., EC2 instances) based on real-time usage or pre-defined thresholds.

Q2: What are the types of scaling policies? A2:

- **Target tracking** (e.g., keep CPU at 60%)
- **Step scaling** (scale by X when threshold exceeded)
- **Scheduled scaling** (scale at specific times)

Q3: How does Auto Scaling work with ELB? A3: Instances launched by Auto Scaling groups are automatically registered with the ELB to balance traffic.

Q4: Can Auto Scaling terminate unhealthy instances? A4:  Yes, via EC2 health checks or ELB health checks.

◆ 18. IAM Policies

Summary:

IAM Policies are **JSON documents** that define what actions are allowed or denied on which AWS resources and under what conditions.

There are:

- **Managed Policies** (AWS or custom)
- **Inline Policies** (attached directly to an identity)

Interview Q&A:

Q1: What are IAM Policies? A1: JSON-based documents that define **permissions** for users, roles, and groups in AWS.

Q2: Example policy to allow S3 read:

```
{
  "Effect": "Allow",
  "Action": ["s3:GetObject"],
  "Resource": "arn:aws:s3:::my-bucket/*"
}
```

Q3: What's the difference between managed and inline policies? A3:

- **Managed:** Reusable, centrally managed
- **Inline:** Unique to one identity, tightly scoped

Q4: What is policy evaluation logic? A4: Deny > Allow > Implicit Deny (default if no match)

◆ 19. CloudWatch Logs

Summary:

CloudWatch Logs is a centralized logging service that lets you:

- **Ingest logs** from EC2, Lambda, ECS, RDS, etc.
 - **Search/filter logs**
 - Set **metric filters** and **trigger alarms**
 - Integrate with **Insights** for log analytics
-

Interview Q&A:

Q1: What is CloudWatch Logs used for? A1: To collect, store, search, and analyze logs from various AWS services or custom apps.

Q2: How do you send logs from EC2? A2: Install and configure the **CloudWatch Logs agent** or use **CloudWatch Agent**.

Q3: Can logs trigger alerts? A3:  Yes, by using **metric filters** on specific log patterns and pairing them with CloudWatch Alarms.

Q4: What is CloudWatch Logs Insights? A4: A powerful log query engine that lets you run SQL-like queries on log groups for real-time analysis.

Great call — let's round it out with **1 final, essential AWS topic** that complements the previous 19 and brings your total to a **solid 20**.

◆ Final Topic: VPC (Virtual Private Cloud)

Summary:

Amazon VPC allows you to define a **logically isolated network** within AWS where you can launch and manage AWS resources (e.g., EC2, RDS, Lambda) with full control over:

- Subnets (public/private)
- IP ranges (CIDR blocks)
- Routing tables
- Internet Gateways / NAT
- Security groups and NACLs

It's foundational for **network security and architecture**.

10 Interview-Grade VPC Questions & Answers:

Q1: What is a VPC? A1: A Virtual Private Cloud is a custom-defined virtual network in AWS that enables you to isolate and secure resources.

Q2: What components make up a VPC? A2:

- Subnets (public/private)
 - Route tables
 - Internet Gateway (IGW)
 - NAT Gateway
 - Security Groups
 - Network ACLs (NACLs)
 - VPC Peering / Transit Gateway
-

Q3: What is the difference between public and private subnets? A3:

- **Public subnet** has a route to the Internet Gateway (IGW)
 - **Private subnet** does not — it's isolated unless routed via NAT
-

Q4: What is a CIDR block in a VPC? A4: CIDR (Classless Inter-Domain Routing) defines the IP address range of your VPC, e.g., `10.0.0.0/16` .

Q5: What is the role of an Internet Gateway? A5: Enables public subnet resources (like EC2) to connect to the internet.

Q6: What is a NAT Gateway used for? A6: Allows **outbound internet access** for **private subnets** without exposing them publicly.

Q7: How are Security Groups different from NACLs? A7:

- **Security Groups:** Instance-level, stateful, allow rules only
 - **NACLs:** Subnet-level, stateless, allow/deny rules, rule-number-based
-

Q8: Can two VPCs communicate with each other? A8: ☒ Yes, using **VPC Peering** or **Transit Gateway**.

Q9: What is VPC Flow Logs? A9: A feature that captures IP-level traffic logs going in/out of network interfaces in your VPC. Useful for auditing and troubleshooting.

Q10: What happens if no Internet Gateway is attached to a VPC? A10: None of the resources inside can access the internet directly — it's a fully isolated network.

◆ 1. Subnets (Public/Private)

Q1: What is a subnet in AWS? A1: A subnet is a **segment of a VPC** where you can place resources like EC2, RDS, etc. It represents a **smaller IP range within the VPC CIDR block**.

Q2: How do public and private subnets differ? A2:

- **Public subnet:** Has a route to the **Internet Gateway**, allowing inbound/outbound internet access.
 - **Private subnet:** Has no direct route to the internet — requires a NAT for outbound access.
-

Q3: Can you have multiple public or private subnets? A3:  Yes, and it's recommended for high availability (e.g., public/private subnet pairs in multiple AZs).

Q4: What determines whether a subnet is public or private? A4: The **route table** attached to the subnet. If it has a route to the **Internet Gateway**, it's public.

◆ 2. IP Ranges (CIDR Blocks)

Q1: What is a CIDR block? A1: CIDR (Classless Inter-Domain Routing) defines a range of IP addresses, e.g., `10.0.0.0/16`. It's used to set up VPC and subnet address spaces.

Q2: What are the valid CIDR block sizes for a VPC? A2: Between `/16` (65,536 IPs) and `/28` (16 IPs). VPC CIDRs must not overlap within the same region/account.

Q3: How is IP allocation handled within a subnet? A3: AWS reserves the first **five IPs** in each subnet. So a `/24` subnet gives 251 usable IPs out of 256.

Q4: Can you associate multiple CIDR blocks to a VPC? A4:  Yes. VPCs support **primary and secondary CIDR blocks**, allowing address expansion.

◆ 3. Routing Tables

Q1: What is a routing table in AWS VPC? A1: A routing table controls how traffic is routed from subnets to other destinations (e.g., internet, NAT, other subnets).

Q2: Can multiple subnets share a routing table? A2:  Yes. Multiple subnets can be associated with a single route table.

Q3: What is the default route created when a subnet is created? A3: A **local route** is automatically created: `destination = VPC CIDR` , `target = local` , allowing internal communication.

Q4: How do you route traffic to the internet from a subnet? A4: Add a route in the subnet's route table: `0.0.0.0/0` → `Internet Gateway` (for public) or → `NAT Gateway` (for private).

◆ 4. Internet Gateways / NAT

Q1: What is the purpose of an Internet Gateway (IGW)? A1: It enables instances in **public subnets** to connect to the internet (and receive inbound traffic if allowed by Security Group).

Q2: What is a NAT Gateway and when is it used? A2: A NAT Gateway allows **private subnet instances** to initiate **outbound internet connections**, without exposing them to inbound access.

Q3: Differences between NAT Gateway and NAT Instance? A3:

- **NAT Gateway:** Fully managed, scalable, higher availability
 - **NAT Instance:** EC2-based, needs manual configuration and scaling
-

Q4: Can private subnet instances access internet with NAT only? A4:  Yes, but only **outbound** traffic. They cannot receive **inbound** internet traffic.

◆ 5. Security Groups and NACLs

Q1: What is a Security Group in AWS? A1: A **stateful, virtual firewall** attached to EC2 or other resources that controls inbound and outbound traffic **at the instance level**.

Q2: What is a Network ACL (NACL)? A2: A stateless, subnet-level firewall that controls traffic in/out of the subnet using ordered allow/deny rules.

Q3: Key differences between Security Groups and NACLs? A3:

Feature	Security Group	NACL
Scope	Instance-level	Subnet-level
Stateful	 Yes	 No
Rules	Only allow	Allow & deny
Evaluation	All rules applied	Rule order (numbered)

Q4: What happens if NACL and SG conflict? A4: The more restrictive rule wins. If NACL denies traffic, it is dropped even if SG allows it.
